

Module -1			*Bloom's Taxonomy Level	Marks
Q.01	a	Define data structures. With a neat diagram, explain the classification of data structures with examples.	L2	5
	b	What do you mean by pattern matching? Outline the Knuth Morris Pratt (KMP) algorithm and illustrate it to find the occurrences of the following pattern. P: ABCDABD S: ABC ABCDAB ABCDABCDABDE	L3	8
	c	Write a program in C to implement push, pop and display operations for stacks using arrays.	L3	7
OR				
Q.02	a	Explain in brief the different functions of dynamic memory allocation.	L2	5
	b	Write functions in C for the following operations without using built-in functions i) Compare two strings. ii) Concatenate two strings. iii) Reverse a string	L3	8
	c	Write a function to evaluate the postfix expression. Illustrate the same for the given postfix expression: ABC-D*+E\$F+ and assume A=6, B=3, C=2, D=5, E=1 and F=7.	L3	7

Module – 1**M****L****C****Q.1****a.**

Define Data Structures. Explain with neat block schematic different type of data structures with examples. What are the primitive operations that can be performed?

10**L2****CO1****b.**

Differentiate between structures and unions shown examples for both.

5**L1****CO1****c.**

What do you mean by pattern matching? Outline knuth, Morris, Pratt pattern matching algorithm.

5**L2****CO1****OR****Q.2****a.**

Define stack. Give the implementation of Push (), POP () and display () functions by considering its empty and full conditions.

7**L2****CO1****b.**

Write an algorithm to evaluate a postfix expression and apply the same for the given postfix expression 6, 2, /, 3, -, 4, 2, *, +

7**L3****CO1****c.**

Write the Postfix form of the following using stack :

6**L3****CO1**

(i) $A*(B*C+D*E) + F$ (ii) $(a + (b*c) / (d-e))$

Module – 1

			M	L	C
Q.1	a.	Define Data Structures. Explain the classification of data structures with a neat diagram.	8	L2	CO1
	b.	Write a C Functions to implement pop , push and display operations for stacks using arrays.	7	L2	CO2
	c.	Differentiate structures and unions.	5	L2	CO1

OR

Q.2	a.	Write an algorithm to evaluate a postfix expression and apply the same for the given postfix expression. 6 2 / 3 - 4 2 * +.	7	L3	CO2
	b.	Explain the dynamic memory allocation function in detail.	8	L2	CO1
	c.	What is Sparse matrix? Give the triplet form of a given matrix and find its transpose $A = \begin{bmatrix} 0 & 0 & 3 & 0 & 4 \\ 0 & 0 & 5 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 6 & 0 & 0 \end{bmatrix}$	5	L3	CO1

Module – 1**M****L****C****Q.1****a.**

Define Data Structures.

04**L1****CO1****b.**

Explain the classification of Data Structures with example.

10**L2****CO1****c.**

Explain all operations of Data Structures.

06**L2****CO1****OR****Q.2****a.**

Explain any five string handling functions supported by 'c' with syntax and example.

10**L2****CO1****b.**

Convert the following infix expression to postfix expression using stack:

$$A + (B * C - (D/E \wedge F) * G) * H$$

10**L3****CO1**